ENTERPRISE OPERATIONS BLUEPRINT

TaskFlow ERP

Complete Architecture, Operations & Delivery Workflow Guide for Executive Leadership and Project Managers

REACT 19 + NODE.JS EXPRESS

MYSQL 24 RELATIONAL TABLES

IST AUTOMATED ENGINE

DYNAMIC RBAC SECURITY

TARGET AUDIENCE

Chief Executive Officer (CEO) & Project Managers (PM)

VERSION & RELEASE

v2.4 Enterprise Production

PRIMARY OBJECTIVE

Operational Transparency, Resource Utilization & Cost Governance

CORE MODULES

18 Integrated Production Engines

1. Executive Summary & Value Proposition

TaskFlow ERP is an all-in-one digital operations platform specifically engineered for software consulting and high-growth technology firms. It eliminates operational friction by uniting project delivery, shift compliance, quality assurance, and financial reconciliations into a single, real-time command center.

MODULES INTEGRATED	BILLING LEAKAGE	PAYROLL GENERATION	TIMEZONE SYNC
18	**0%**	**1-Click**	**IST**
ZERO SILOS	AUDIT VERIFIED	AUTO-COMPUTED	ASIA/KOLKATA

Stakeholder RACI Matrix (Roles & Responsibility Scope)

Every feature and data endpoint in TaskFlow is strictly gated through dynamic Role-Based Access Control (RBAC):

Role	Core Responsibilities	System Scope	Primary Impact
CEO / Super Admin	Company strategy, shifts & holiday calendar, financial approvals, invoice auditing, payroll dispatch, organizational oversight.	100% Read & Write across all 24 database entities.	PROFIT & GOVERNANCE
Project Manager	Client onboarding, project scoping, milestone planning, task & subtask delegation, timesheet approvals, QA defect triage.	Assigned projects, team members, timesheets & sprints.	ON-TIME DELIVERY
Developer / Staff	Shift clock-in/out, daily task execution, timesheet hour logging, break compliance, leave & expense reimbursement filing.	Personal portal, self-service timesheets & tasks.	PRODUCTIVE OUTPUT
QA Engineer	Test cases creation, executing test runs, logging defect tickets with reproduction steps, verifying bug fixes.	Project testing suites, bug reports & quality gates.	DEFECT ELIMINATION

Key Takeaway for CEO: You possess real-time visibility into project profitability, employee attendance punctuality, and client billing rates without having to request manual Excel reports from managers.

Key Takeaway for Project Managers: You can assign tasks with or without immediate team allocation, verify whether timesheets match actual attendance hours, and guarantee high code quality prior to client deployment.

2. Granular RBAC Permission & Action Matrix

TaskFlow enforces strict dynamic permission checking across 4 distinct operational vectors for all 18 modules: **Read (Navigation/View)**, **Create (+ Add Button)**, **Update (Edit & Lifecycle Actions)**, and **Delete**.

Module / System Function	CEO / Super Admin	Project Manager	Developer / Staff	QA Engineer
Project Management (`project_management`)	ALL (C,R,U,D)	C, R, U	READ (ASSIGNED)	READ (ASSIGNED)
Task & Subtasks (`task_management`)	ALL (C,R,U,D)	ALL (C,R,U,D)	R, U (STATUS)	READ ONLY
Timesheet Management (`timesheet_management`)	ALL (AUDIT)	R, U (APPROVE)	C, R (OWN LOG)	C, R (OWN LOG)
QA & Bug Tracking (`qa_management`)	ALL (C,R,U,D)	R, U (REVIEW)	R (FIX BUGS)	ALL (C,R,U,D)
Client Accounts & Invoicing (`client_reports`)	ALL (C,R,U,D)	R, C (INVOICES)	⦸ HIDDEN	⦸ HIDDEN
Payroll & Salary Runs (`payroll_management`)	ALL (C,R,U,D)	⦸ HIDDEN	⦸ HIDDEN	⦸ HIDDEN
Resource Utilization (`resource_management`)	ALL (C,R,U,D)	ALL (ALLOCATE)	⦸ HIDDEN	⦸ HIDDEN
IT Asset Inventory (`asset_management`)	ALL (C,R,U,D)	R (TEAM ASSETS)	R (OWN ASSET)	R (OWN ASSET)
Attendance & Shift Punch (`attendance_management`)	ALL (CONFIGURE)	R, U (REVIEW)	C, R (SELF PUNCH)	C, R (SELF PUNCH)
Leave Requests (`leave_management`)	ALL (POLICY)	R, U (APPROVE)	C, R (APPLY)	C, R (APPLY)
Expense Claims (`expense_management`)	ALL (DISBURSE)	R, U (APPROVE)	C, R (CLAIM)	C, R (CLAIM)
Roles & System Permissions (`role_management`)	ALL (FULL SETUP)	⦸ HIDDEN	⦸ HIDDEN	⦸ HIDDEN
Staff & User Directory (`staff_management`)	ALL (C,R,U,D)	READ ONLY	⦸ HIDDEN	⦸ HIDDEN
Team Chat & Notice Board (`chat_management`)	ALL (BROADCAST)	ACTIVE (CHAT/POST)	ACTIVE (CHAT/READ)	ACTIVE (CHAT/READ)

> **Zero Data Leakage Guarantee:** When a user lacks `read` permission for any module, it is automatically purged from the Navigation Sidebar and blocked at the React Router layer. Sensitive administrative actions like adding roles, editing salary structures, or viewing client billings are impossible for non-authorized personnel.

2. Core Business Workflows

A. Project Delivery & Execution Lifecycle

How a project transitions from client contract to completion and billing:

1. **Client & Project Creation (CEO / PM)**
   A client profile is created with currency and billing terms. A project is initiated (Name, Description, Target Deadlines). *Note: Assigning team members is optional at inception.*

2. **Work Breakdown Structure (PM)**
   Project roadmap is decomposed into granular **Tasks** and **Subtasks** with assigned priorities (Critical, High, Medium, Low), estimated hours, and assigned engineers.

3. **Task Execution & Live Updates (Developers)**
   Engineers move task states (`Pending` ➔ `In Progress` ➔ `Ready for QA`) and link their Git branches/commits.

4. **Daily Timesheet Filing & Verification (Developer ➔ PM)**
   Engineers record hours spent on tasks tagged as **Billable** or **Non-billable**. PM verifies submitted hours against daily punch duration and clicks **Approve**.

5. **Quality Assurance & Bug Regression (QA Engineer)**
   QA runs test cases. If a defect occurs, a **Bug Ticket** is filed and reassigned to developer. Once resolved, QA marks it `Verified Complete`.

6. **Invoicing & Financial Realization (CEO / Accounts)**
   Approved billable hours are compiled with 1-click into branded PDF invoices and sent to the client. Realized revenue is tracked on the CEO dashboard.

3. Workforce Governance & Shift Architecture

TaskFlow features an automated IST (Asia/Kolkata) attendance engine that enforces workplace discipline, eliminates manual timekeeping, and automatically feeds into the monthly payroll module.

Automated Shift Timeline & Penalty Rules

Time Window (IST)	Event / Trigger	System Action & Compliance Rule	Attendance Status
09:30 AM – 10:00 AM	Early Arrival Window	Clock-in allowed up to 30 mins before official shift start. Employee selects *Office* or *WFH*.	ON TIME
10:00 AM – 10:30 AM	Official Grace Period	Standard 30-minute grace window for daily commuting variations.	PRESENT (GRACE)
10:31 AM – 12:00 PM	Late Arrival	Registers a **Late Instance**. Up to 3 late instances allowed per month. Exceeding 3 instances automatically deducts a First Half Day.	LATE FLAG
After 12:00 PM	Excessive Delay	Automatic conversion to **Half-Day** presence regardless of past late quota.	HALF DAY
During Shift	Break Governance	Lunch Break (40m) & Tea Break (20m). Timed with live stopwatch on dashboard. Overages flagged as Excess Breaks.	MONITORED
19:00 PM Onwards	Shift Completion	Employee clocks out. System verifies net hours worked (Total elapsed minus excess breaks).	COMPLETED
11:00 PM Auto-Lock	Forgotten Punch Out	Shift auto-terminated. Employee cannot clock in next morning without submitting a **Second Half Regularization Leave**.	AUTO-CLOSED

Leave Management & Quota Structure

Employees are allocated fixed yearly allowances. Every leave follows an automated approval chain:

- **Casual Leave (CL – 12 Days):** Planned personal time off. Requires prior PM approval.
- **Sick Leave (SL – 8 Days):** Health emergencies; medical certificates attachable.
- **Paid Leave (PL – 15 Days):** Earned leave accumulated across project service.
- **Short Leave (2 Hours):** Allows stepping away for urgent matters without forfeiting half-day pay.

4. Comprehensive 18-Module Functional Breakdown

An executive overview of all 18 modules, detailing user inputs, internal logic, and business deliverables:

1. 🔐 Dynamic RBAC & Authentication

SECURITY

INPUTS & CONTROLS
Bcrypt hashed credentials, JWT bearer tokens, dynamic permission toggle table.

CEO & PM DELIVERABLE
Guarantees sensitive revenue, salary, and client data is strictly shielded from unauthorized staff.

2. ⏰ Attendance & Shift Management

OPERATIONS

INPUTS & CONTROLS
Web punch-in, IP / WFH location check, IST timestamping, late count aggregator.

CEO & PM DELIVERABLE
Automated punctuality enforcement, elimination of "buddy punching", instant export of monthly times.

3. 🧉 Break & Productivity Compliance

PRODUCTIVITY

INPUTS & CONTROLS
Lunch timer (40 min cap), Tea timer (20 min cap), simultaneous break locks.

CEO & PM DELIVERABLE
Prevents unaccounted team downtime during active project sprints.

4. 📁 Project Management & Milestones

DELIVERY

INPUTS & CONTROLS
Client assignment, milestone schedules, hourly budgets, status flags (Active, Hold, Done).

CEO & PM DELIVERABLE
High-level visibility into project burn rate, delivery milestones, and deadline tracking.

5. 📋 Task & Subtask Delegation Engine

EXECUTION

INPUTS & CONTROLS
Kanban/List view, Priority (Critical to Low), due dates, estimated hours, subtask checklists.

CEO & PM DELIVERABLE
Granular delegation of work packages with real-time sprint status tracking.

6. ⏱️ Timesheet & Billing Reconciliations

FINANCE

INPUTS & CONTROLS
Daily task hour logging, billable vs non-billable tagging, PM approval trigger.

CEO & PM DELIVERABLE
Ensures 100% of billable client engineering hours are audited, approved, and billed without loss.

4. Comprehensive 18-Module Breakdown (Contd.)

7. 🏖️ Leave Management & Approval Chains

HR OPERATIONS

INPUTS & CONTROLS
CL, SL, PL, Comp-off balances; Full, Half, or Short leave request options.

CEO & PM DELIVERABLE
Eliminates unapproved absenteeism; auto-deducts unpaid leave from monthly payroll.

8. 💰 Expense Management & Reimbursements

ACCOUNTING

INPUTS & CONTROLS
Expense category (Travel, Client, SaaS), bill receipt uploads, multi-stage approval.

CEO & PM DELIVERABLE
Tight policy compliance on corporate expenditure with transparent audit trails.

9. 🧪 QA Test Case & Bug Tracking Engine

QUALITY

INPUTS & CONTROLS
Test case suites, execution logs (Pass/Fail), bug severity, reproduction steps.

CEO & PM DELIVERABLE
Directly measures software stability and prevents untested code from reaching clients.

10. 🏢 Client Accounts & Automated Invoicing

REVENUE

INPUTS & CONTROLS
Client directory, agreed billing rates, project linkages, tax calculation.

CEO & PM DELIVERABLE
1-Click generation of professional client invoices populated from approved timesheets.

11. 💵 Automated Payroll & Salary Dispatches

COMPENSATION

INPUTS & CONTROLS
CTC, Basic, HRA, Allowances, PF, Professional Tax, Unpaid leave deductions.

CEO & PM DELIVERABLE
Zero-error monthly payroll generation and automated payslip PDF distribution.

12. 💻 IT Hardware & Software Asset Registry

ASSET MANAGEMENT

INPUTS & CONTROLS
Serial numbers, asset tags, laptop specifications, allocation history, return checklist.

CEO & PM DELIVERABLE
Guarantees accountability for company laptops, monitors, and cloud license seats.

13. 📢 Notice Board & Mandatory Acknowledgments

CORPORATE HR

INPUTS & CONTROLS
Broadcast announcements, priority pinning, mandatory "Read & Acknowledge" button.

CEO & PM DELIVERABLE
Proof of policy communication for HR compliance and team-wide memos.

14. 🗓️ Official Company Holiday Calendar

CALENDAR

INPUTS & CONTROLS
Annual gazetted, state, and optional holiday scheduling.

CEO & PM DELIVERABLE
Auto-exempts employees on official holidays to preserve accurate attendance scores.

15. 💬 Real-Time Team Chat & Collaboration

INTERNAL COMMS

INPUTS & CONTROLS
Sub-millisecond Socket.io engine, direct 1-on-1 chats, project team channels, files.

CEO & PM DELIVERABLE
Keeps project conversations centralized inside TaskFlow without third-party tools.

16. 🎫 IT Helpdesk & Infrastructure Tickets

SUPPORT

INPUTS & CONTROLS
Hardware issue tickets, VPN/credential requests, priority queue, resolution status.

CEO & PM DELIVERABLE
Minimizes development downtime caused by infrastructure or workstation failures.

17. 🏠 Work From Home (WFH) Governance

REMOTE WORK

INPUTS & CONTROLS
Remote request submissions, manager approval workflow, punch widget remote toggle.

CEO & PM DELIVERABLE
Disciplined remote work framework with full manager approval control.

18. 📈 Performance Appraisals & Star Ratings

TALENT & GROWTH

INPUTS & CONTROLS
Structured 1-5 star ratings across delivery, code quality, communication, and ownership.

CEO & PM DELIVERABLE
Data-driven appraisal history to reward top performers and identify training needs.

5. Executive Dashboards & Decision Metrics

TaskFlow transforms raw database entries into actionable intelligence for executive management:

For the Chief Executive Officer (CEO)

Executive Metric	Source Module	Strategic Business Action
Organization Billable Ratio	Timesheets & Project Budgets	Measure engineering utilization against total payroll cost.
Outstanding Invoices & Aging	Client Accounts & Invoicing	Optimize accounts receivable and cash flow cycles.
Monthly Payroll Liabilities	Payroll & Statutory Deductions (PF/PT/TDS)	Audit monthly salary overheads vs corporate revenue.
Attendance & Retention Trends	Attendance, Leaves & Appraisals	Identify workforce stability, burn-out patterns, or chronic tardiness.

For the Project Manager (PM)

Operational Metric	Source Module	Tactical Delivery Action
Sprint Task Velocity & Burndown	Tasks & Subtask Delegation	Ensure critical roadmap milestones remain on schedule.
Timesheet Compliance Rate	Timesheet Approvals	Prevent delayed billing by ensuring daily submission by 19:30 IST.
Defect Leakage & Bug Severity	QA Test Cases & Bug Reports	Halt releases if Critical/Major defects remain unresolved.
Team Capacity & Planned Leaves	Leave Management & WFH Requests	Forecast sprint capacity accurately before committing client deliverables.

TaskFlow ERP Operations Directorate

Enterprise Edition • Verified Architecture Specification

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